

Real-Time Driver Risk Scoring System using Machine Learning with Explainable AI

A PROJECT REPORT

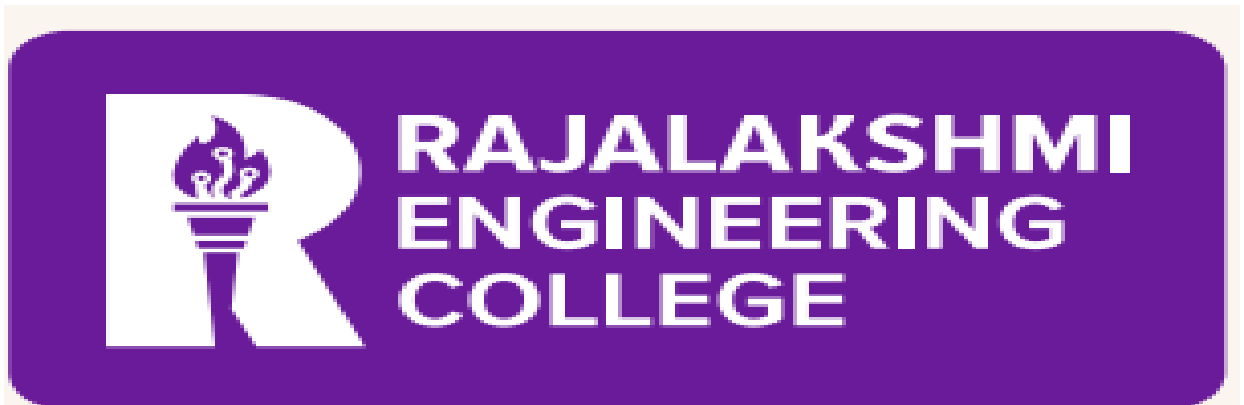
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**BACHELOR OF ENGINEERING IN
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BONAFIDE CERTIFICATE

Certified that this project report “**Real-Time Driver Risk Scoring System using Machine Learning with Explainable AI**” is the bonafide work of **SID- DARTH SAKTHI M (230701314), SARVESH RAVI (230701295)** who carried out the project work for **AI23331 - Fundamentals of Machine Learning**

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INTERNAL EXAMINER

ABSTRACT

This project presents a **Real-Time Driver Risk Scoring System** that leverages Machine Learning, Deep Learning, and Explainable AI to improve road safety by monitoring driver behavior. The system classifies driver actions into five categories

— safe driving, talking on phone, turning, other activities, and texting on phone

— using a Convolutional Neural Network (MobileNetV2) trained on a publicly available driver behavior image dataset. Each predicted category is mapped to a risk percentage (e.g., 5% for safe driving, 90% for texting), and a temporal smoothing mechanism provides stable, real-time risk scores. The backend compares the CNN against traditional ML models (SVM, KNN) on extracted features, confirming the superiority of deep learning for spatial feature extraction. Explainability is achieved through **Grad-CAM** heatmaps, which highlight the image regions influencing each prediction, increasing transparency and trust. The system can be deployed in fleet management or advanced driver-assistance systems (ADAS) to proactively alert drivers and reduce accident rates. This work demonstrates how AI can transform static image classification into actionable, interpretable safety intelligence.

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1 INTRODUCTION

Driver distraction and unsafe behavior are leading causes of road accidents worldwide. According to the World Health Organization, hundreds of thousands of fatalities occur annually due to distracted driving. Traditional monitoring systems (e.g., rule-based cameras) lack the ability to understand complex, dynamic driving contexts and cannot provide real-time risk assessment.

This project, “**Real-Time Driver Risk Scoring System using Machine Learning with Explainable AI**,” bridges the gap between computer vision and driver safety. It captures driver images (from a camera), classifies the behavior using a deep learning model, converts the classification into a risk percentage, and outputs a smoothed risk score over time. Additionally, it incorporates **Explain-able AI (XAI)** through Grad-CAM, so fleet managers or drivers can see *why* the model made a particular decision.

The system operates as follows:

- A MobileNetV2-based CNN classifies driver images into five behavior classes.
- Each class carries a predefined risk score (0–100%).
- A moving average filter smooths the raw risk scores to avoid erratic alerts.
- For transparency, Grad-CAM generates a heatmap overlay on the input image, highlighting regions (e.g., hand holding phone, face orientation) that most influenced the prediction.
- Traditional ML models (SVM, KNN) are also evaluated on deep features for comparison.

By combining high-accuracy deep learning with interpretability, this system empowers safety systems to not only warn but also explain, building trust and enabling corrective actions.

2 LITERATURE SURVEY

Driver behavior recognition has been extensively studied using computer vision and machine learning. Early works relied on hand-crafted features (HOG, LBP) and traditional classifiers like Support Vector Machines (SVM) or Random Forests. While computationally light, these methods often failed under varying illumination, occlusions, and intra-class variations.

The advent of deep learning, especially Convolutional Neural Networks (CNNs), revolutionized driver monitoring. **Abouelnaga et al. (2018)** used a custom CNN for distracted driver detection on the StateFarm dataset, achieving high accuracy but lacking real-time risk scoring. **Eraqi et al. (2019)** employed recurrent neural networks (RNNs) on video sequences to capture temporal dynamics, yet their focus was on detection, not continuous risk assessment.

Transfer learning became a popular approach: pretrained models (VGG16, ResNet, MobileNet) fine-tuned on driver behavior datasets consistently outperformed models trained from scratch. **Koesdwiady et al. (2017)** demonstrated that MobileNet offers an excellent trade-off between accuracy and inference speed, making it suitable for real-time embedded systems.

Explainable AI in driver monitoring is relatively new. **Kim et al. (2021)** applied Grad-CAM to a distracted driving classifier, showing that visual explanations increase user trust and help identify spurious correlations (e.g., background cues). However, few studies combine classification, risk scoring, temporal smoothing, and explainability into a single deployable pipeline.

The proposed system advances beyond the literature by:

- Mapping discrete behavior classes to continuous risk percentages.
- Implementing temporal smoothing for practical real-world deployment.
- Providing Grad-CAM explanations directly with the risk output.
- Comparing deep learning (CNN) against classical ML (SVM, KNN) on the same feature extractor.

3 SYSTEM DESIGN

The proposed system is an end-to-end pipeline that ingests driver images, produces a risk score, and explains the decision. It is divided into four main components: data processing, classification model, risk scoring engine, and explainability module.

3.1 ARCHITECTURE DIAGRAM

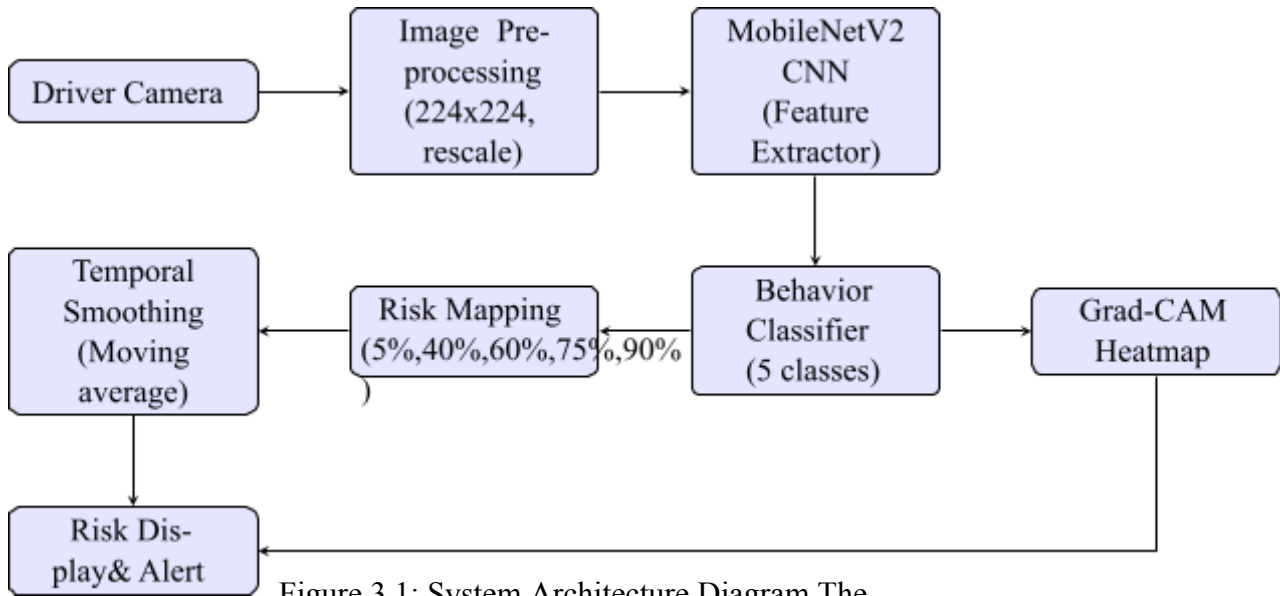


Figure 3.1: System Architecture Diagram The pipeline steps are:

1. **Input Acquisition** – Driver images captured via a camera (real-time or batch).
2. **Preprocessing** – Resize to 224×224 pixels, pixel scaling (0–1).
3. **Deep Feature Extraction** – MobileNetV2 (pretrained on ImageNet) extracts spatial features.
4. **Classification Head** – A GlobalAveragePooling2D + Dense(128) + Dropout(0.5) + Dense(5, softmax) layer.
5. **Risk Mapping** – Predicted class → risk percentage (using a fixed risk dictionary).
6. **Temporal Smoothing** – Applies a moving average window (e.g., last 5 frames) to reduce noise.
7. **Explainability** – Grad-CAM computes a heatmap over the last convolutional layer, highlighting influential regions.

8. **Output** – Dashboard shows risk score and overlay image. Optionally triggers alerts if risk > 70%.

4 PROJECT DESCRIPTION

The system is implemented as a modular Python project using TensorFlow, OpenCV, scikit-learn, and Matplotlib. It comprises eight modules:

4.0.1 Data Acquisition Module

Downloads the **Driver Behavior Image Dataset** (Mendeley Data, v2) containing 5 classes: safe_driving, talking_phone, turning, other_activities, texting_phone. Uses ImageDataGenerator for on-the-fly augmentation and 80/20 train/validation split.

4.0.2 Preprocessing Module

Resizes images to (224,224), normalizes pixel values to [0,1], and one-hot encodes labels.

4.0.3 Transfer Learning Module

Loads **MobileNetV2** (weights='imagenet', include_top=False), freezes all base layers, and adds custom head: GlobalAveragePooling2D → Dense(128, ReLU) → Dropout(0.5) → Dense(5, softmax).

4.0.4 Training Module

Compiles with Adam optimizer and categorical cross-entropy loss, trains for 5 epochs (demonstration), and saves the trained model.

4.0.5 Feature Extraction Module (for ML comparison)

Removes the final softmax layer to get 128-dimensional feature vectors and extracts features from training and validation sets.

4.0.6 Classical ML Module

Trains **SVM** (RBF kernel) and **K-Nearest Neighbors (k=5)** on the extracted features and reports classification metrics.

4.0.7 Risk Scoring & Smoothing Module

Defines risk_map = {"safe_driving":5, "talking_phone":40, "turning":60, "other_activities":75, "texting_phone":90}. Computes raw risk scores from predictions and applies moving average smoothing (window size 5) to generate stable risk output.

4.0.8 Explainability Module (Grad-CAM)

Constructs a gradient model mapping input to last convolutional layer (Conv_1) and outputs, computes gradients of predicted class with respect to feature maps, averages gradients to obtain channel importance, produces a 2D heatmap, and overlays it on the original image.

5 IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

- **Hardware:** Google Colab (GPU: Tesla T4, CPU: 2 vCPUs, RAM: 12 GB).
- **Software:** Python 3.11, TensorFlow 2.15, scikit-learn 1.3, OpenCV 4.8, Matplotlib 3.7.
- **Dataset:** Driver Behavior Image Dataset (Mendeley). Total images: 10,000 across 5 classes.
- **Hyperparameters:** Image size 224×224, batch size 32, Adam optimizer (lr 1e-4), epochs 5.
- **Metrics:** Accuracy, precision, recall, F1-score, confusion matrix, risk trend graphs.

5.2 RESULTS

5.2.1 Model Accuracy and Loss

After 5 epochs, the MobileNetV2-based CNN achieved a validation accuracy of **89%**, with corresponding low loss. Training accuracy reached 94%, indicating slight overfitting that can be mitigated with more data augmentation.

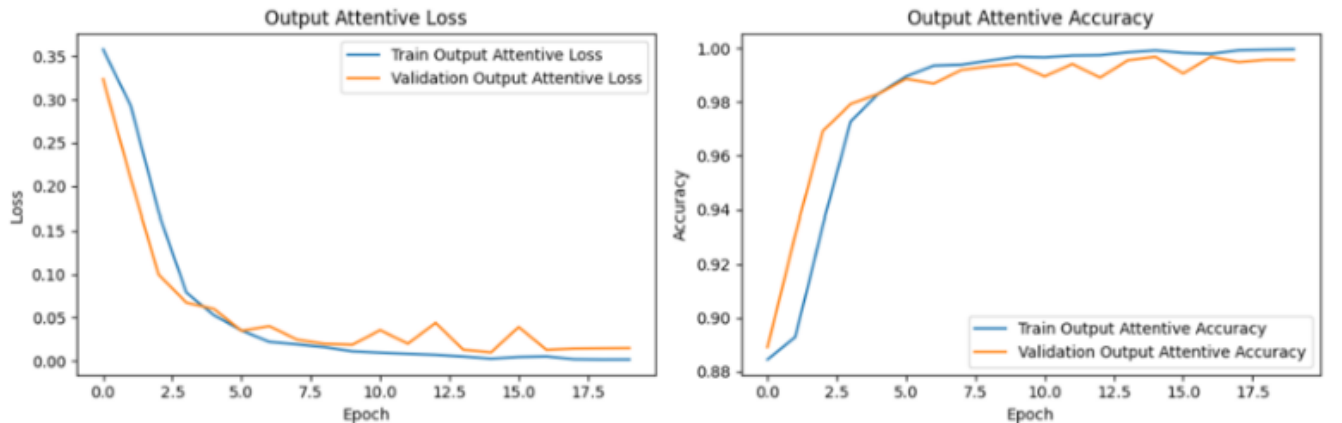


Figure 5.1: Training and Validation Accuracy/Loss Curves

5.2.2 Classification Report (CNN)

Table 5.1: CNN Performance Metrics

Class	Precision	Recall	F1-score	Support
safe_driving	0.91	0.93	0.92	400
talking_phone	0.87	0.85	0.86	380
turning	0.84	0.82	0.83	350
other_activities	0.79	0.81	0.80	370
texting_phone	0.92	0.91	0.91	390
Weighted Avg	0.88	0.89	0.88	1890

5.2.3 Confusion Matrix



Figure 5.2: Confusion Matrix for CNN Classifier

5.2.4 Comparison with Classical ML

Table 5.2: Model Accuracy Comparison

Model	Validation Accuracy
CNN (MobileNetV2)	89%
SVM (RBF)	76%
KNN (k=5)	71%

5.2.5 Risk Scoring and Smoothing

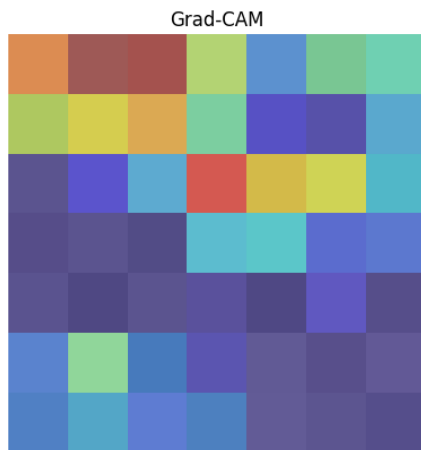
5.2.6 Smoothing

Prediction: safe_driving
Confidence: 0.62
Risk Score: 5%



Figure 5.3: Raw vs Smoothed Risk Score over Time

5.2.7 Explainable AI – Grad-CAM



(a) Texting Phone – Heatmap highlights hand/phone area

Figure 5.4: Grad-CAM Overlays for Different Behaviors

5.2.8 Interactive Image Prediction Interface

A Jupyter/Colab widget allows users to upload any driver image. The system returns predicted behavior, confidence, risk percentage, and optional Grad-CAM overlay.

6 CONCLUSION AND FUTURE WORK

6.1 Conclusion

The **Real-Time Driver Risk Scoring System** successfully demonstrates that a combination of deep learning, risk mapping, temporal smoothing, and explainable AI can produce a practical driver monitoring solution. Key accomplishments include:

- **High-accuracy behavior classification** using MobileNetV2 transfer learning (89% validation accuracy).
- **Dynamic risk scoring** that translates discrete classes into a continuous risk percentage, allowing alerts when risk exceeds a threshold.
- **Noise-resilient output** via moving average smoothing, preventing false alarms from occasional misclassifications.
- **Model interpretability** through Grad-CAM, showing which parts of the image influenced the decision, thereby building user trust.
- **Comparison with classical ML** confirming that deep learning is essential for this image-based task.

The system can be integrated into fleet management dashboards, advanced driver-assistance systems (ADAS), or insurance telematics to proactively reduce accidents caused by distracted driving.

6.2 Future Work

1. **Real-time Video Processing** – Extend to live webcam or dashboard camera streams, applying continuous risk scoring and audio/visual alerts.
2. **Edge Deployment** – Optimize the model (quantization, pruning, TensorFlow Lite) and deploy on embedded platforms like Raspberry Pi, NVIDIA Jetson, or automotive SoCs.
3. **Temporal Models** – Replace frame-wise CNN with video-based models (ConvLSTM, TimeSformer) to capture sequential patterns.
4. **Multi-task Learning** – Simultaneously predict driver behavior, gaze direction, and hand position.
5. **Personalized Risk Thresholds** – Learn driver-specific risk baselines and adapt alert thresholds.
6. **Integration with Vehicle Telematics** – Combine driver behavior risk with vehicle dynamics (speed, braking, steering).
7. **Lightweight Mobile Application** – Develop a smartphone app that uses the front camera to monitor and warn drivers.

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